

School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Chebyshev's Inequality, Chernoff Bounds, and Jensen's Inequality

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Chebyshev's Inequality, Chernoff Bounds, and Jensen's Inequality

Overview

The lecture covers probabilistic tail bounds including Markov, Chebyshev, and Chernoff bounds, followed by Jensen's Inequality and its connection to convexity and expectations.

Definitions and Notation

Markov Inequality:

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Chebyshev Inequality:

$$P(|X - \mu| \geq a) \leq \frac{\text{Var}(X)}{a^2}$$

Random Variable (Binomial):

$$X \sim \text{Binomial}(n, \frac{1}{2}), \quad \mu = \frac{n}{2}, \quad \text{Var}(X) = \frac{n}{4}$$

Convex Function Definition:

A function $g(x)$ is convex on interval S if for any $x_1, x_2 \in S$ and $\alpha \in [0, 1]$:

$$g(\alpha x_1 + (1 - \alpha)x_2) \leq \alpha g(x_1) + (1 - \alpha)g(x_2)$$

Expectation:

$$\mathbb{E}[X] = \sum p_X(x_i) x_i$$

Expectation of function:

$$\mathbb{E}[g(X)] = \sum p_X(x_i) g(x_i)$$

Main Results

Chebyshev's Inequality

Uses mean + variance. Controls deviation from mean.

$$P(|X - \mu| \geq a) \leq \frac{\text{Var}(X)}{a^2}$$

Proof Idea:

Apply Markov on $(X - \mu)^2$:

$$P(|X - \mu| \geq a) = P((X - \mu)^2 \geq a^2) \leq \frac{\mathbb{E}[(X - \mu)^2]}{a^2} = \frac{\text{Var}(X)}{a^2}$$

Running Example (Binomial)

$$P\left(X \geq \frac{3}{4}n\right) = P\left(|X - \frac{n}{2}| \geq \frac{n}{4}\right) \leq \frac{n}{(n/4)^2} = \frac{4}{n}$$

Key Insight: Bound decreases as n increases.

Real-Life Example

Mean = 70, Std deviation = 10

$$P(|X - 70| \geq 30) \leq \frac{10^2}{30^2} = \frac{1}{9}$$

At most $\frac{1}{9}$ students are far from mean.

Solved Example 1

$$\mu = 70, \sigma = 5 \Rightarrow \sigma^2 = 25$$

$$X \geq 90 \Rightarrow |X - 70| \geq 20$$

$$P(X \geq 90) \leq P(|X - 70| \geq 20) \leq \frac{25}{400} = \frac{1}{16} = 0.0625$$

Result: Much tighter than Markov.

Solved Example 2 (Network Latency)

$$\mu = 50, \sigma^2 = 16$$

$$P(|X - 50| \geq 20) \leq \frac{16}{400} = \frac{1}{25}$$

Result: At most 4% packets outside range.

Chernoff Bound: Key Idea

Progression:

$$\text{Markov} \rightarrow X, \quad \text{Chebyshev} \rightarrow (X - \mu)^2, \quad \text{Chernoff} \rightarrow e^{tX}$$

Key Insight: Exponential grows very fast. Tail events get amplified.

Core Trick

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

Apply Markov:

$$P(e^{tX} \geq e^{ta}) \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}}$$

Final Chernoff Bound

$$P(X \geq a) \leq \min_{t>0} \left\{ \frac{\mathbb{E}[e^{tX}]}{e^{ta}} \right\}$$

Key Insight: Choose best t for tightest bound. Exponential decay instead of polynomial.

Lower Tail Chernoff

$$P(X \leq a) \leq \min_{t<0} \left\{ \frac{\mathbb{E}[e^{tX}]}{e^{ta}} \right\}$$

Poisson Chernoff Bound

$$\mathbb{E}[e^{tX}] = \sum_{i=0}^{\infty} e^{ti} \frac{e^{-\lambda} \lambda^i}{i!} = e^{\lambda(e^t-1)}$$

$$P(X \geq a) \leq \min_{t>0} e^{\lambda(e^t-1)-ta}$$

Minimized at:

$$t = \ln \left(\frac{a}{\lambda} \right)$$

Thus:

$$P(X \geq a) \leq e^{a-\lambda} \left(\frac{\lambda}{a} \right)^a$$

Chernoff Example (Binomial Lower Tail)

$$X \sim \text{Binomial}(20, 0.5), \quad \mu = 10, \quad \delta = 1$$

$$P(X \leq 0) \leq \exp \left(-\frac{\mu \delta^2}{2} \right) = e^{-5} \approx 0.0067$$

Jensen's Inequality

We know:

$$\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$$

General result:

$$\mathbb{E}[X^a] \geq (\mathbb{E}[X])^a$$

Convexity

$$g''(x) \geq 0$$

From Convexity to Expectation

$$g\left(\sum \alpha_i x_i\right) \leq \sum \alpha_i g(x_i)$$

Interpretation: α_i act like probabilities.

$$g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)]$$

Jensen's Theorem

If $g(x)$ is convex and $X \in S$:

$$g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)]$$

Jensen Simple Example

$$X = \{1, 3\}, \quad P = 0.5$$

$$\mathbb{E}[X] = 2$$

$$g(\mathbb{E}[X]) = 2^2 = 4$$

$$\mathbb{E}[g(X)] = 0.5(1^2) + 0.5(3^2) = 5$$

Thus:

$$g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)]$$

Jensen Solved Example

$$X = \{2, 4\}, \quad \mathbb{E}[X] = 3$$

$$g(\mathbb{E}[X]) = 9$$

$$\mathbb{E}[g(X)] = 0.5(4) + 0.5(16) = 10$$

Derivations / Proofs

Chebyshev Proof

$$P(|X - \mu| \geq a) = P((X - \mu)^2 \geq a^2) \leq \frac{\mathbb{E}[(X - \mu)^2]}{a^2} = \frac{\text{Var}(X)}{a^2}$$

Chernoff Proof Structure

Transform:

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

Apply Markov and minimize over t .

Jensen Proof Idea

$$g\left(\sum \alpha_i x_i\right) \leq \sum \alpha_i g(x_i)$$

Replace weights with probabilities to obtain expectation form.

Worked Examples

- Exam Scores $\rightarrow \frac{1}{9}$ bound
- Grades variance $\rightarrow \frac{1}{16}$
- Network latency $\rightarrow \frac{1}{25}$
- Binomial Chernoff $\rightarrow e^{-5} \approx 0.0067$
- Jensen verification $\rightarrow 4 \leq 5$

Summary

Markov:

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Chebyshev:

$$P(|X - \mu| \geq a) \leq \frac{\text{Var}(X)}{a^2}$$

Chernoff:

$$P(X \geq a) \leq \min_{t>0} \frac{\mathbb{E}[e^{tX}]}{e^{ta}}$$

Jensen:

$$g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)]$$

Key Ideas:

- Tail bounds improve from Markov $\rightarrow$ Chebyshev $\rightarrow$ Chernoff
- Exponential gives strongest bounds
- Convexity connects to expectation via Jensen